FINITENESS OF THE FIBRE PRODUCT OF TWO FINITE MODULES OVER A FIBRE PRODUCT OF RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- The Situation of a fibre product of rings and the identification of the two base changes of the fibre product module with the outer modules. the author cites Lemma lemma-module-over-fibre-product, counted by the sibling card S1-algebra-module-over-fibre-product at 643-715, at proof lines 774-776. Supplies the identifications used to lift the generators.
- Surjectivity of the structure map from the fibre product of rings onto each factor and the identification of its kernel with the given ideal. used at proof lines 782-790 and 798-802. Lets coefficients in the ideal be lifted to elements of the fibre product.
- The fibre product of modules consists of pairs agreeing in the common quotient. used at proof lines 794-802 to recognise the difference of two elements as a pair vanishing in the first component.. Supplies the data and identifications used in the corresponding step of the proof.

Source reference. Original lemma 0D2H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16382–16382. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9879–9879. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A, A', B, B', I, M, M', N, \varphi$ be as in Lemma 07RU. If N finite over B and M' finite over A', then $N' = N \times_{\varphi, M} M'$ is finite over B'.*

Proof. We will use the results of Lemma 07RU without further mention. Choose generators $y_1, \ldots, y_r$ of N over B and generators $x_1, \ldots, x_s$ of M' over A'. Using that $N = N' \otimes_{B'} B$ and $B' \to B$ is surjective we can find $u_1, \ldots, u_r \in N'$ mapping to $y_1, \ldots, y_r$ in N. Using that $M' = N' \otimes_{B'} A'$ we can find $v_1, \ldots, v_t \in N'$ such that $x_i = \sum v_j \otimes a'_{ij}$ for some $a'_{ij} \in A'$. In particular we see that the images $\overline{v}_j \in M'$ of the v_j generate M' over A'. We claim that $u_1, \ldots, u_r, v_1, \ldots, v_t$ generate N' as a B'-module. Namely, pick $\xi \in N'$. We first choose $b'_1, \ldots, b'_r \in B'$ such that ξ and $\sum b'_i u_i$ map to the same element of N. This is possible because $B' \to B$ is surjective and $y_1, \ldots, y_r$ generate N over B. The difference $\xi - \sum b'_i u_i$ is of the form $(0, \theta)$ for some θ in IM'. Say θ is $\sum t_j \overline{v}_j$ with $t_j \in I$. As $J = \mathrm{Ker}(B' \to B)$ maps isomorphically to I we can choose $s_j \in J \subset B'$ mapping to t_j. Because $N' = N \times_{\varphi, M} M'$ it follows that $\xi = \sum b'_i u_i + \sum s_j v_j$ as desired. $\square$